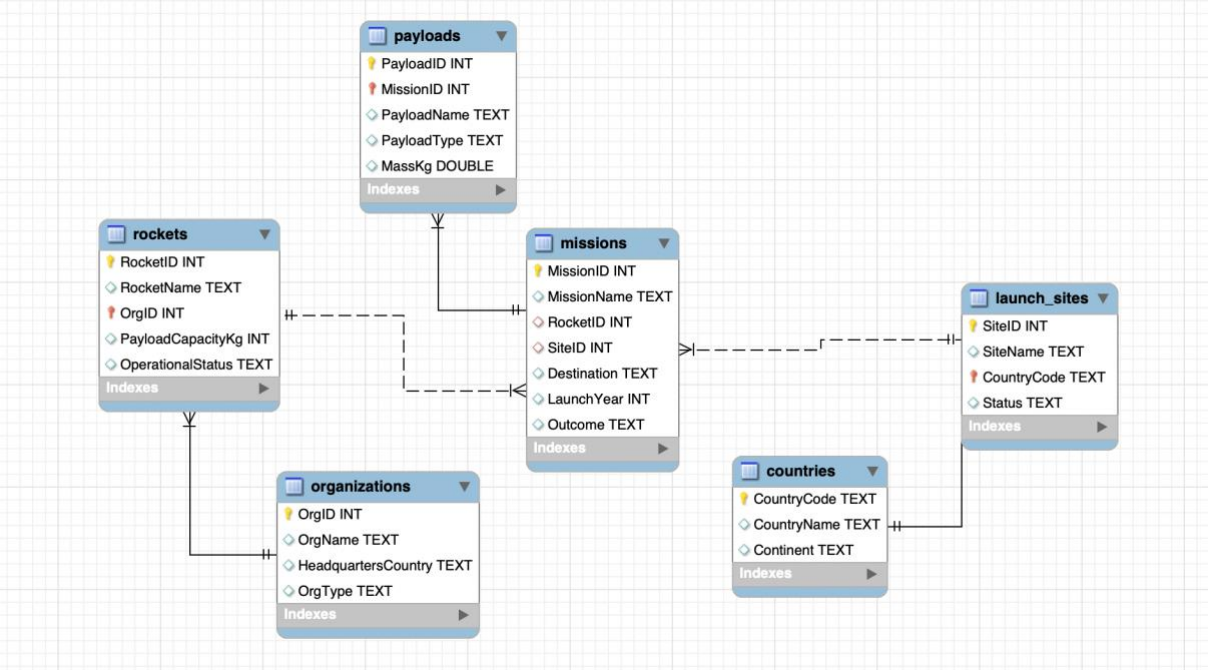
Dr. Forouraghi

CSC 351 D01

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Homework #4

EER Diagram



SQL Problems

```sql
-- 1) Retrieve the mission name, rocket name, organization name, launch site name, destination,
-- launch year, and outcome for all space missions without using table aliases. Sort the results
-- in descending order by launch year and ascending order by mission name.

select missions.MissionName,
    rockets.RocketName,
    organizations.OrgName,
    launch_sites.SiteName,
    missions.Destination,
    missions.LaunchYear,
    missions.Outcome
from missions natural join rockets
               natural join organizations
               natural join launch_sites
order by missions.LaunchYear desc, missions.MissionName asc;
```

MissionName	RocketName	OrgName	SiteName	Destination	LaunchYear	Outcome
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	2024	Success
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

```sql
-- 2) Retrieve the mission name, rocket name, payload name, payload type, and payload mass (in kg)
-- for all payloads launched in 2024 without using table aliases. Sort the results in descending order
-- by payload mass.

select missions.MissionName,
    rockets.RocketName,
    payloads.PayloadName,
    payloads.PayloadType,
    payloads.MassKg
from missions natural join payloads
               natural join rockets
where missions.LaunchYear = 2024
order by payloads.MassKg desc;
```

MissionName	RocketName	PayloadName	PayloadType	MassKg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100

```sql
-- 3) Retrieve all launch sites along with their matching mission names, rockets, and launch years
-- for missions that took place in 2024 without using table aliases, ensuring that launch sites with
-- no recorded 2024 missions still appear in the results with null values for the mission fields.
-- Sort the results in alphabetical order by launch site name.

select launch_sites.SiteName,
    missions.MissionName,
    rockets.RocketName,
    missions.LaunchYear
from launch_sites left join missions on launch_sites.SiteID = missions.SiteID and missions.LaunchYear = 2024
                left join rockets using (RocketID)
order by launch_sites.SiteName asc;
```

SiteName	MissionName	RocketName	LaunchYear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

```sql
-- 4) Write a SQL query using a Common Table Expression (CTE) to calculate the total payload mass (in kg)
-- deployed by each organization across all missions without using table aliases. Retrieve the organization
-- name, headquarters country, organization type, and total payload mass, and sort the results in descending
-- order by total payload mass.

with total_payload_mass as(
    select organizations.OrgName,
        sum(payloads.MassKg) as sum_masskg
    from organizations natural join rockets
                        natural join missions
                        natural join payloads
    group by organizations.OrgName
)
select organizations.OrgName,
    organizations.HeadquartersCountry,
    organizations.OrgType,
    total_payload_mass.sum_masskg
from organizations natural join total_payload_mass
order by total_payload_mass.sum_masskg desc;
```

OrgName	HeadquartersCoun...	OrgType	sum_masskg
National Aeronautics and Space Administration	USA	Government	25800
Space Exploration Technologies Corp	USA	Private	13692
Japan Aerospace Exploration Agency	JPN	Government	6000
Roscosmos State Corporation	RUS	Government	2450
China National Space Administration	CHN	Government	1200
Indian Space Research Organisation	IND	Government	150
Rocket Lab	NZL	Private	100

```sql
-- 5) Write a SQL query using two separate Common Table Expressions (CTEs)—one to calculate total missions
-- by country code and another to calculate total payload mass by country code without using table aliases.
-- Then, combine these CTEs with the countries table to display the country name, total number of missions,
-- and total payload mass (in kg). Sort the results in descending order by total payload mass.

with total_missions as(
    select count(missions.MissionID) as mission_count,
        countries.CountryCode
    from countries natural join launch_sites
                    natural join missions
    group by countries.CountryCode
),

total_payload_mass as(
    select sum(payloads.MassKg) as sum_masskg,
        countries.CountryCode
    from countries natural join launch_sites
                    natural join missions
                    natural join payloads
    group by countries.CountryCode
)

select countries.CountryName,
        total_missions.mission_count,
        total_payload_mass.sum_masskg
from countries natural join total_missions
                natural join total_payload_mass
order by total_payload_mass.sum_masskg desc;
```

CountryName	mission_count	sum_masskg
United States	3	39492
Japan	1	6000
Russia	1	2450
China	1	1200
India	1	150
New Zealand	1	100